

OutboundAI — End-to-End Project Documentation

Outbound + Inbound AI Voice Calling Platform

Prepared for Ladder Hub / AiCallingDemo-Pro

Version: Inbound Test Branch Documentation

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Purpose

This document explains the full OutboundAI project: setup, settings, features, day-to-day usage, testing, troubleshooting, and how to safely modify or deploy the system. It is written for a beginner-friendly workflow using Coolify, GitHub, Supabase, LiveKit, Vobiz, Cloudflare R2/S3, Gemini, and the web dashboard.

Important Safety Notes

- Do not share API keys, Supabase service role keys, LiveKit secrets, Vobiz credentials, Cloudflare R2 secrets, or admin passwords.
- Do not deploy experimental inbound changes directly to the live domain until all tests pass on the inbound-test domain.
- Keep the live main branch stable. New inbound work should continue on the inbound-calls branch until production-ready.

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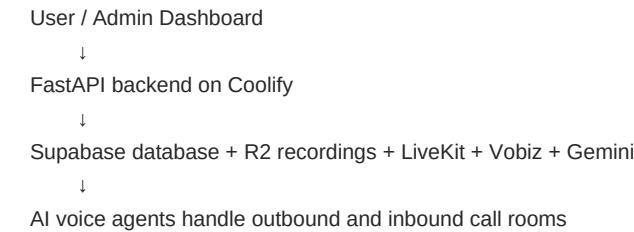
1. Project Overview

OutboundAI is an AI voice calling platform that supports outbound calling, inbound call answering, CRM lead management, call logs, appointment booking, recording storage, and admin dashboard controls.

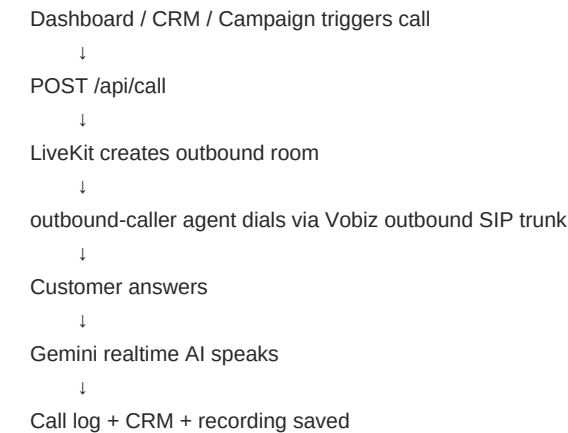
Module	Purpose	Current Status
FastAPI backend	Provides REST APIs for dashboard, calls, CRM, auth, exports, settings, recordings, and inbound setup.	Working
LiveKit agents	Runs AI voice workers for outbound-caller and inbound-agent.	Working after separate worker ports
Vobiz SIP	Provides outbound/inbound SIP trunking and phone number routing.	Outbound working; inbound routing configured
Supabase	Stores settings, CRM leads, call logs, appointments, error logs, and status data.	Working; Phase 6.1 columns required
Cloudflare R2 / S3	Stores call recordings and supports retention cleanup.	Configured
Dashboard UI	Admin dashboard for calls, CRM, logs, recordings, inbound setup, and settings.	Working on inbound-test

2. Architecture and Call Flows

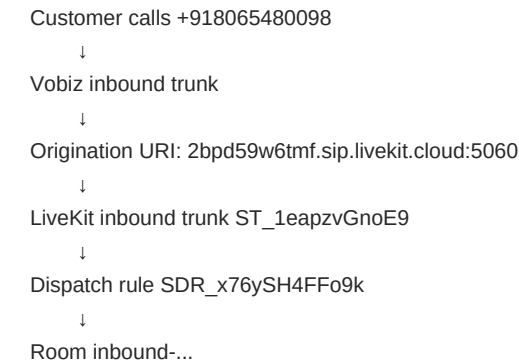
2.1 High-Level Architecture



2.2 Outbound Call Flow



2.3 Inbound Call Flow





inbound-agent joins, reads caller phone, checks CRM, speaks, logs call

3. Repositories, Branches, and Deployment Structure

Item	Value / Recommendation
GitHub repository	karuna0001/AiCallingDemo-Pro
Live production branch	main
Inbound test branch	inbound-calls
Live production domain	voice.ladderaiaagent.com
Inbound test domain	inbound-test.ladderaiaagent.com
Coolify live app	AiCallingDemo-Pro
Coolify inbound test app	AiCallingDemo-Pro-Inbound-Test
Important rule	Do not merge inbound-calls into main until all tests pass.

4. Environment Variables and Settings

4.1 Core Environment Variables

Variable	Purpose	Example / Notes
PORT	FastAPI web server port	8000
LIVEKIT_URL	LiveKit project WebSocket URL	wss://...livekit.cloud
LIVEKIT_API_KEY	LiveKit API key	Secret; do not share
LIVEKIT_API_SECRET	LiveKit API secret	Secret; do not share
GOOGLE_API_KEY	Gemini API key	Secret; do not share
GEMINI_MODEL	Realtime Gemini model	gemini-3.1-flash-live-preview
GEMINI_TTS_VOICE	Gemini TTS voice	Aoede
USE_GEMINI_REALTIME	Enable realtime voice model	true
SUPABASE_URL	Supabase project URL	https://...supabase.co
SUPABASE_SERVICE_KEY	Supabase service role key	Secret; must be correct
ADMIN_USERNAME	Dashboard login username	admin or custom
ADMIN_PASSWORD	Dashboard login password	Strong password
SESSION_SECRET	Signed cookie secret	Long random string

4.2 SIP / Vobiz / LiveKit Variables

Variable	Purpose	Notes
OUTBOUND_TRUNK_ID	LiveKit outbound trunk for placing calls	Do not reuse for inbound
VOBIZ_SIP_DOMAIN	Vobiz SIP domain	From Vobiz dashboard
VOBIZ_USERNAME	Vobiz SIP username	Secret
VOBIZ_PASSWORD	Vobiz SIP password	Secret
VOBIZ_OUTBOUND_NUMBER	Outbound caller ID / number	Use approved Vobiz number
DEFAULT_TRANSFER_NUMBER	Human transfer number	Example: +918065480098

4.3 Recording Storage Variables

Variable	Purpose	Notes
S3_ACCESS_KEY_ID	R2/S3 access key	Secret
S3_SECRET_ACCESS_KEY	R2/S3 secret	Secret
S3_ENDPOINT_URL	Cloudflare R2 endpoint	https://ACCOUNT_ID.r2.cloudflarestorage.com
S3_REGION	R2 region	auto
S3_BUCKET	Recording bucket name	Example: aicall-recordings
RECORDING_AUTO_DELETE_ENABLED	Auto cleanup old recordings	true / false
RECORDING_RETENTION_DAYS	Days to keep recordings	7

RECORDING_CLEANUP_TIME	Daily cleanup time	02:00
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4.4 Inbound Settings Stored in Supabase

Setting	Recommended Value	Purpose
INBOUND_ENABLED	true	Enable inbound dashboard/agent behavior
INBOUND_PHONE_NUMBER	+918065480098	Number customers call
INBOUND_TRUNK_ID	ST_1eapzvGnoE9	LiveKit inbound trunk ID
INBOUND_DISPATCH_RULE_ID	SDR_x76ySH4FFo9k	LiveKit dispatch rule ID
INBOUND_AGENT_PROFILE_ID	inbound-agent	Agent profile used by inbound calls
INBOUND_BUSINESS_NAME	Ladder Hub	Name AI says during inbound greeting
INBOUND_SERVICE_TYPE	Ai Voice Call Agent	Service context
INBOUND_GREETING_MESSAGE	Hi, thank you for calling Ladder Hub. How can I help you today?	Inbound greeting
INBOUND_FAQ_TEXT	Business FAQ text	Used to answer inbound questions
INBOUND_AFTER_HOURS_MODE	false	Optional after-hours logic
DEFAULT_TRANSFER_NUMBER	+918065480098	Human transfer destination

5. Supabase Database and SQL Setup

Run the project `supabase_schema.sql` whenever new phases add columns. For inbound support, these `call_logs` columns are required:

```
ALTER TABLE call_logs ADD COLUMN IF NOT EXISTS call_type text DEFAULT 'outbound';
ALTER TABLE call_logs ADD COLUMN IF NOT EXISTS room_name text;
ALTER TABLE call_logs ADD COLUMN IF NOT EXISTS livekit_call_id text;
ALTER TABLE call_logs ADD COLUMN IF NOT EXISTS sip_trunk_id text;
ALTER TABLE call_logs ADD COLUMN IF NOT EXISTS sip_dispatch_rule_id text;
ALTER TABLE call_logs ADD COLUMN IF NOT EXISTS trunk_phone_number text;
ALTER TABLE call_logs ADD COLUMN IF NOT EXISTS transferred_to text;
ALTER TABLE call_logs ADD COLUMN IF NOT EXISTS transfer_reason text;
```

```
CREATE INDEX IF NOT EXISTS idx_call_logs_call_type ON call_logs(call_type);
CREATE INDEX IF NOT EXISTS idx_call_logs_phone_type ON call_logs(phone_number, call_type);
CREATE INDEX IF NOT EXISTS idx_call_logs_timestamp_type ON call_logs(timestamp, call_type);
```

Table	Used For
settings	Dashboard and system configuration values
crm_contacts	CRM leads, caller identity, CRM status, notes, follow-up
call_logs	Outbound and inbound call records, duration, outcome, SIP metadata
appointments	Appointments booked during calls
error_logs	Dashboard-visible logs and troubleshooting messages
crm_statuses	Custom CRM statuses and colors

6. Dashboard Features and How to Use Them

Tab	What It Does	Daily Use
Stats	Shows call totals, booking rate, live config, charts	Check system health and activity
Single Call	Place one outbound test or sales call	Use for quick testing
Batch Call	Call a small list of numbers	Use for small batches
Campaigns	Organize outbound calling by campaign	Use for structured sales campaigns
Agents	Manage/view agent configuration	Use only if needed
AI Prompt	Edit default/custom AI behavior	Modify carefully; test after every change
Appointments	View appointments booked by AI	Use for follow-up
Call Logs	Review calls, outcomes, recordings, reasons	Daily review

CRM	Add, upload, filter, download, call selected leads	Main lead management area
Inbound	Inbound setup, settings, health, call logs, stats	Manage inbound calling
Settings	Recording storage, retention, system settings	Admin-only
Logs	Live app logs from error_logs/server	Troubleshooting
Setup	Outbound setup and helper tools	Initial setup / admin

7. Outbound Calling Workflow

- Use Single Call for testing one number.
- Use CRM Call Selected Leads for controlled calling from uploaded leads.
- Use Campaigns for campaign-style calling.
- Outbound calls use OUTBOUND_TRUNK_ID and must not use INBOUND_TRUNK_ID.

Step	Action	Expected Result
1	Enter customer number and business/service details	Call request accepted
2	Backend creates LiveKit outbound room	Room starts with call-...
3	outbound-caller agent dials through Vobiz	Customer phone rings
4	AI speaks and handles conversation	Appointment/callback/outcome captured
5	Call ends	Call log and recording are saved

8. Inbound Calling Workflow

- Customer calls +918065480098.
- Vobiz inbound trunk forwards call to LiveKit SIP endpoint.
- LiveKit dispatches the call to an inbound room and assigns inbound-agent.
- inbound-agent reads SIP caller number, checks CRM, starts AI session, and logs the call.

Component	Configured Value
Vobiz Primary URI	2bpd59w6tmf.sip.livekit.cloud:5060
Vobiz linked number	+918065480098
LiveKit inbound trunk	ST_1eapzvGnoE9
LiveKit dispatch rule	SDR_x76ySH4FFo9k
Agent name	inbound-agent
Room prefix	inbound

9. CRM and Lead Management

The CRM can store manual leads, uploaded leads, Google Sheet imported leads, outbound call leads, and inbound callers. New inbound callers are created as Unknown Caller with source inbound_call if not already found.

Feature	How to Use	Expected Result
Manual Add Lead	CRM tab → fill fields → Add Lead	Lead appears in CRM
Upload Leads	Upload CSV/XLSX with phone_number and lead_name	Bulk leads imported
Filters	Use city/source/status/outcome/date filters	CRM list updates
Downloads	Download filtered Excel or CSV	Filtered lead file downloads
Call Selected Leads	Select leads → Start Calling Selected Leads	Only selected leads called
Status Manager	Add/delete status, mark hot/not interested	CRM status updated

10. Recordings and R2/S3 Storage

- Recordings are stored in Cloudflare R2 or S3-compatible storage.
- Recording metadata can be saved in call_logs: recording_url, recording_object_key, recording_size_bytes, recording_deleted, recording_deleted_at.

- Retention cleanup can delete old recordings when enabled and confirmed.

Endpoint	Purpose
GET /api/recordings/storage	View storage usage and recording summary
POST /api/recordings/cleanup	Delete old recordings only with confirmation
Cleanup confirmation	{"confirm": "DELETE_OLD_RECORDINGS"}

11. LiveKit and Vobiz Configuration

11.1 LiveKit

- Use separate trunk IDs for outbound and inbound.
- Outbound uses OUTBOUND_TRUNK_ID.
- Inbound uses INBOUND_TRUNK_ID and INBOUND_DISPATCH_RULE_ID.
- Inbound dispatch rule should use agent_name inbound-agent and room prefix inbound.

11.2 Vobiz

- Outbound trunk is for AI placing calls to customers.
- Inbound trunk is for customer calls routed to LiveKit.
- Origination URI should be LiveKit SIP endpoint without sip: prefix and with port 5060 if Vobiz requires domain:port.

Vobiz Item	Value
Inbound Trunk Name	AIInboundDemo
Primary Origination URI	2bpd59w6tmf.sip.livekit.cloud:5060
Transport	TCP
Linked Number	+918065480098
Call Recording	Enabled

12. Deployment with Coolify

Deployment	Branch	Domain	Purpose
AiCallingDemo-Pro	main	voice.ladderaiaagent.com	Stable live outbound app
AiCallingDemo-Pro-Inbound-Test	inbound-calls	inbound-test.ladderaiaagent.com	Testing outbound + inbound before merge

12.1 Coolify Build Settings

Setting	Value
Build Pack	Dockerfile
Base Directory	/
Dockerfile Location	/Dockerfile
Port Exposes	8000
Health URL	/api/health

12.2 Worker Startup

FastAPI runs on PORT=8000

outbound-caller LiveKit worker health port = 8081

inbound-agent LiveKit worker health port = 8082

13. Testing Checklist

Area	Test	Pass Criteria
Health	Open /api/health	status ok, services configured
Login	Login/logout/wrong password	Auth works
Outbound	Single call to your phone	AI speaks, call log saves
Batch	2-3 numbers	No crashes, logs saved

Campaign	Create/start/pause campaign	Calls work and logs save
CRM	Add/upload/filter/download leads	Leads visible, export correct
Appointment	Ask AI to book tomorrow 5 PM	Appointment appears
Recording	Complete one call	Recording link saves
Inbound Health	Inbound tab badges	All major badges green
Inbound Call	Call +918065480098	AI answers inbound call
Inbound CRM	Call from new number	Unknown Caller created
Inbound Stats	Refresh inbound stats	Count increases
Logs	Check Coolify/app logs	No crash, no 8081 conflict

14. Troubleshooting Guide

Problem	Likely Cause	Fix
Dashboard shows Invalid API key	Wrong Supabase service key	Paste correct SUPABASE_SERVICE_KEY and redeploy
Inbound stats/calls fail with missing column	Supabase SQL not updated	Run Phase 6.1 ALTER TABLE SQL
Container fails: address already in use 8081	Both workers using same LiveKit health port	Use outbound 8081 and inbound 8082
Inbound call creates room but AI does not speak	Agent job crash or Gemini session not starting	Check Coolify logs after call
CRM lookup crash: result.data None	Supabase maybe_single returned None	Harden get_crm_contact_by_phone and fallback to Unknown Caller
Vobiz URI rejects sip: prefix	Vobiz expects domain:port	Use 2bpd59w6tmf.sip.livekit.cloud:5060
Outbound call does not ring	Outbound trunk or Vobiz credential issue	Check OUTBOUND_TRUNK_ID and Vobiz SIP credentials
Recording not saved	R2/S3 env wrong or bucket missing	Check S3 endpoint, keys, bucket, region auto

15. How to Modify the Project Safely

15.1 Git Workflow

```
cd "E:\AiCallingDemo-Pro"
git switch inbound-calls
git status
# Make changes
py -m py_compile server.py db.py agent.py tools.py prompts.py
git add <changed files>
git commit -m "Clear message"
git push origin inbound-calls
```

15.2 Safe Modification Rules

- Never edit live main branch directly for experimental inbound changes.
- Always compile before commit.
- Only redeploy inbound-test while testing inbound changes.
- Do not add patch files to commits unless intentionally needed.
- Do not print secret values in /api/health or dashboard badges.

15.3 Common Files

File	Purpose	Modify Carefully When
server.py	FastAPI endpoints, auth, setup routes, dashboard APIs	Adding API endpoints or changing auth
agent.py	LiveKit agent endpoint and voice session	Changing inbound/outbound AI behavior
tools.py	AI function tools like appointments, CRM, transfer	Changing what AI can do during calls
db.py	Supabase helpers and CRUD logic	Changing schema/data access
prompts.py	Prompt building and AI instructions	Changing AI personality/logic

ui/index.html	Dashboard frontend	Changing tabs, forms, buttons
start.sh	Container startup, workers, FastAPI	Changing worker startup or ports
requirements.txt	Python dependencies	Adding libraries
supabase_schema.sql	Database schema	Adding columns/tables/indexes

16. Go-Live and Maintenance SOP

16.1 Go-Live Rule

- Only merge inbound-calls into main after all checklist items pass on inbound-test.
- Before merge, export/download call logs and confirm no test secrets are exposed.
- After merge, redeploy live only during low traffic time.

16.2 Daily Maintenance

Daily Task	Action
Check health	Open dashboard Stats and /api/health
Check logs	Review Logs tab for crashes/errors
Check CRM	Review new inbound_call leads and Hot Leads
Check appointments	Confirm appointments are valid and followed up
Check recordings	Verify storage usage and retention
Test one call	Do one inbound or outbound test after any redeploy

16.3 Backup / Recovery

- Keep GitHub branch history clean with small commits.
- Use Supabase backups if available before major schema changes.
- Keep environment variable values stored securely outside the codebase.
- If deployment fails, rollback in Coolify to previous successful deployment.

Appendix A — Useful URLs and Commands

Item	Value
Test dashboard	https://inbound-test.ladderaiaagent.com
Test health	https://inbound-test.ladderaiaagent.com/api/health
Inbound stats	https://inbound-test.ladderaiaagent.com/api/inbound/stats
Inbound calls	https://inbound-test.ladderaiaagent.com/api/inbound/calls
Live dashboard	https://voice.ladderaiaagent.com
Inbound number	+918065480098

Compile check

```
py -m py_compile server.py db.py agent.py tools.py prompts.py
```

Git status and push

```
git status
```

```
git add <files>
```

```
git commit -m "Your message"
```

```
git push origin inbound-calls
```

Common health tests

```
curl https://inbound-test.ladderaiaagent.com/api/health
```

```
curl https://inbound-test.ladderaiaagent.com/api/inbound/stats
```